

HSC Mathematics Assessment Task 3

Term 2, 2009

Question 1 (12 Marks)**Marks**

- a) Evaluate the expression $e^5 + \log_e 50$ correct to 2 decimal places, 2
- b) Find the value of $\log_a(bc)^2$ given that $\log_a b = 2.75$ and $\log_a c = 0.25$. 2
- c) Each year a person's life seems only $\frac{9}{10}$ as long as the previous year from the 2
second year of their life. What is the oldest a person could expect to feel, assuming
they could live forever?
- d) Show that the curve $y = e^{x^2}$ is concave up for all values of x . 3
- e) Differentiate $y = \frac{e^x + 1}{2x}$ with respect to x . 2
- f) Find $\int (1+7x)^6 dx$. 1

Question 2 (12 Marks)**Start a new page**

- a) Consider the curve $y = \frac{1}{x^2}$ in the first quadrant. 1
i) Write an expression for x in terms of y for this function. 1
ii) Find the area in the first quadrant between $y = \frac{1}{x^2}$, the y axis and the lines 3
 $y = 4$ and $y = 2$ (leaving your answer in surd form).
- b) Use the trapezoidal rule with 4 trapezia to calculate an approximation for 4
 $\int_0^4 \frac{1}{x^2 + 1} dx$.
- c) Consider the function $y = e^{kx}$ 1
i) Find the second derivative of this function. 1
ii) Find all possible values of k for which $y = e^{kx}$ satisfies the equation 3

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 35y = 0$$

Question 3 (12 Marks)[Start a new page](#)

a) Find $\int (3 + e^x)^2 dx$ **2**

b) Find $\int_1^4 \sqrt{x} dx$ **2**

c) A river 40 metres wide is measured for depth every 10 metres directly across its width.

These measurements, from bank to bank are given in the following table.

Distance from bank	0	10	20	30	40
Depth in metres	0	12.1	17.2	6.9	2

i) Find an approximation for the cross-sectional area of the river, using **4**
Simpson's rule and 5 function values

d) A solid is generated when the region in the first quadrant enclosed between the
curve $y = x^2$, the y axis and the line $y = 4$ is rotated about the x axis.

i) draw a sketch of this information showing the region with shading. **1**

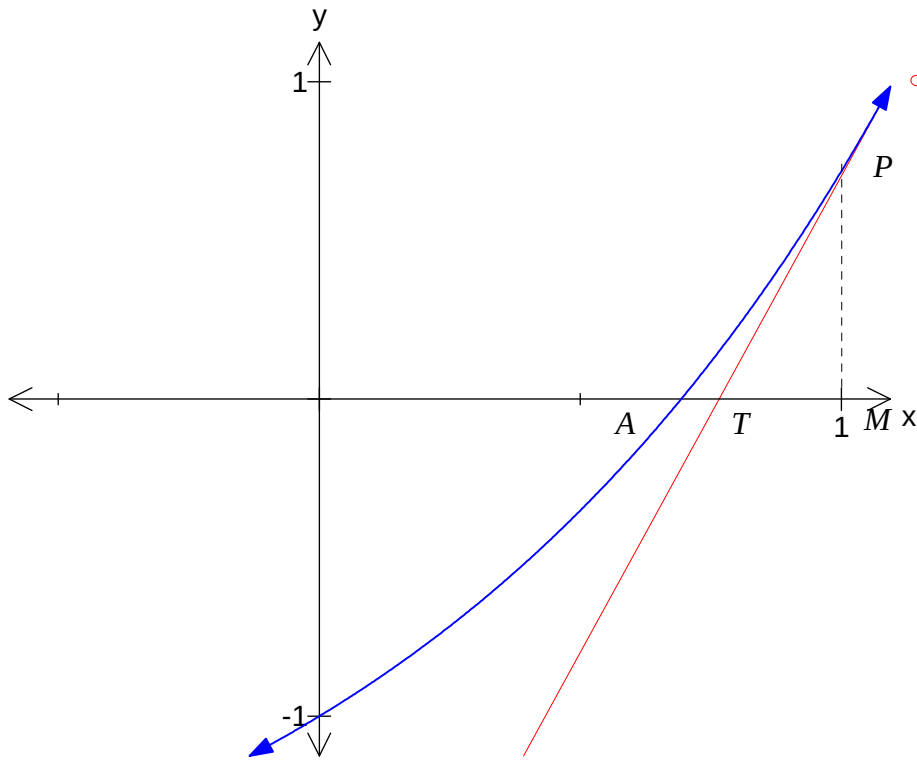
ii) Find the volume of the solid formed by rotating this region (leaving your **3**
answer in terms of π .)

Question 4 (12 Marks) **Start a new page**

a) A geometric series has its n th term given by $T_n = (x-2)^n$.

- i. Write out the first 3 terms of this series without expansion. **1**
- ii. Find the range of values of x for which the series has a limiting sum. **1**
- iii. Find this sum in terms of x in its simplest form. **2**

b) The diagram shows the graph of the function $y = e^x - 2$



The curve $y = e^x - 2$ cuts the x axis at A.

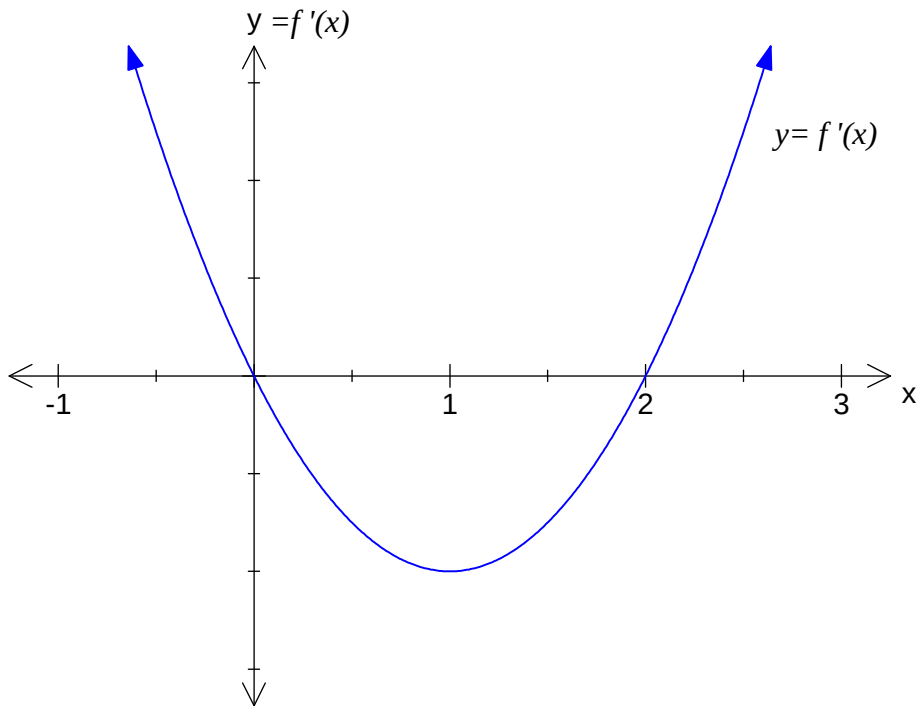
P is the point $(1, e-2)$ on the curve.

The tangent to the curve at P cuts the x axis at T

- i. Find the x coordinate of the point A (in exact form). **1**
- ii. Find the equation of the tangent at P and hence the x coordinate of the point T. **3**
- iii. Find the area enclosed by the curve, the tangent at P and the x axis. **4**

Start a new page

- a) The diagram shows a sketch of the gradient function of the curve $y = f(x)$



It is known that this gradient function passes through the point $(1, -3)$

- i) Show that $y = f'(x)$ has equation $f'(x) = 3x^2 - 6x$ 2
- ii) Find a possible equation for this curve and sketch its graph. 2

- b) Robyn invests \$1200 in an extra superannuation fund paying 4% pa compound interest.

At the beginning of the next year she deposits \$600 into this account which also earns 4% pa compound interest and continues to add \$600 to this account at the beginning of each year for a period of time (interest compounding annually at 4% pa).

- | | | |
|------|---|---|
| i. | What will be the value of Robyn's first 2 deposits when the first deposit has been invested for n years? | 2 |
| ii. | Derive an expression for the total accumulated amount of all her deposits at the end of n years (expressed in simplest form). | 4 |
| iii. | Robyn plans to withdraw all her funds when their total value reaches at least \$25000. How many deposits will she make in total including the first \$1200? | 2 |

Question 1

$$a) \quad 150 \cdot 0.225$$

$$\approx 150 \cdot 0.2$$

$$b) \quad 2 \log_a(bc)$$

$$= 2(\log_a b + \log_a c)$$

$$= 2(2.75 + 0.25)$$

$$= 2 \times 3$$

$$= 6$$

$$c) \quad S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1}{1-\frac{9}{10}}$$

$$= \frac{1}{\frac{1}{10}}$$

$$= 10$$

10 years old.

$$d) \quad y = e^{x^2}$$

$$\frac{dy}{dx} = 2x e^{x^2}$$

$$\frac{d^2y}{dx^2} = 2x \cdot 2x e^{x^2} + e^{x^2} \cdot 2$$

$$= 2(2x^2 + 1) e^{x^2}$$

Concave up $\frac{d^2y}{dx^2} > 0$

$2e^{x^2} > 0$ for all x

$2x^2 + 1 > 0$ for all x

$\therefore$ Curve Concave up for all x

$$e) \quad \frac{dy}{dx} = \frac{2xe^{x^2} - (e^{x^2} + 1)2}{(2x)^2}$$

$$= \frac{2xe^{x^2} - 2e^{x^2} - 2}{4x^2}$$

$$f) \quad \frac{(1+7x)^7}{7 \times 7} + c$$

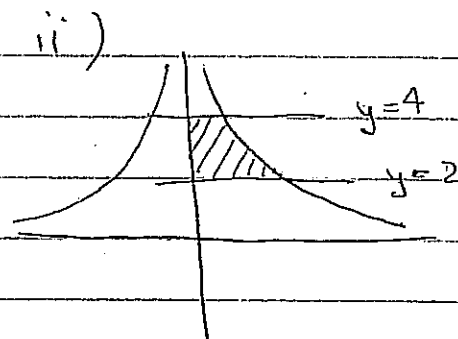
$$= \frac{(1+7x)^7}{49} + c$$

Question 2

$$a) \quad i) \quad y = x^{-2}$$

$$\frac{dy}{dx} = -2x^{-3}$$

$$x = y^{-\frac{1}{2}}$$



$$A = \int_2^4 y^{-\frac{1}{2}} dy$$

$$= \left[2y^{\frac{1}{2}} \right]_2^4$$

$$= \left[2\sqrt{y} \right]_2^4$$

$$= 2\sqrt{4} - 2\sqrt{2}$$

$$= 4 - 2\sqrt{2}$$

Area is $4 - 2\sqrt{2}$ unit

Question 2

b) $h = \frac{4-0}{4}$
 $= 1$

x	0	1	2	3	4
y	1	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{17}$

$$A \doteq \frac{h}{2} \{ y_1 + y_5 + 2(y_2 + y_3 + y_4) \}$$

$$= \frac{1}{2} \left\{ 1 + \frac{1}{17} + 2\left(\frac{1}{2} + \frac{1}{5} + \frac{1}{10}\right) \right\}$$

$$= \frac{1}{2} \left\{ \frac{1}{17} + 2\left(\frac{4}{5}\right) \right\}$$

$$= 1 \frac{28}{85}$$

c) $y = e^{kx}$
 $\frac{dy}{dx} = k e^{kx}$

$$\frac{d^2y}{dx^2} = k^2 e^{kx}$$

$$\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - 35y = 0$$

$$k^2 e^{kx} + 2k e^{kx} - 35 e^{kx} = 0$$

$$e^{kx} (k^2 + 2k - 35) = 0$$

$$[e^{kx} > 0] \quad (-k+7)(k+5) = 0$$

$$k = -7, 5$$

Question three

$$\begin{aligned} \text{a)} \quad & \int (3+e^x)^2 dx \\ &= \int 9 + 6e^x + (e^x)^2 dx \\ &= \int 9 + 6e^x + e^{2x} dx \\ &= 9x + 6e^x + \frac{1}{2}e^{2x} + C \end{aligned}$$

$$\begin{aligned} \text{b)} \quad & \int_1^4 \sqrt{x} dx = \int_1^4 x^{\frac{1}{2}} dx \\ &= \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^4 \\ &= \frac{2}{3} [x\sqrt{x}]_1^4 \\ &= \frac{2}{3} (8-1) \\ &= \frac{14}{3} \end{aligned}$$

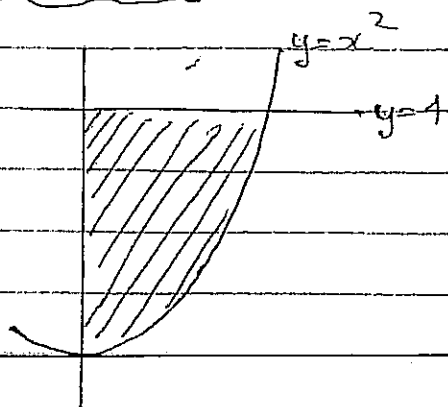
$$\begin{aligned} \text{c)} \quad & h = 10 \\ & A \approx \frac{h}{3} \{ (y_1 + y_5) + 2(y_3) + 4(y_2 + y_4) \} \\ &= \frac{10}{3} \{ (0 + 2) + 2(17 \cdot 2) + 4(12 \cdot 1 + 6 \cdot 9) \} \\ &= \frac{10}{3} \{ 2 + 34 \cdot 4 + 76 \} \end{aligned}$$

$$= 374 \cdot 6$$

Area is $374 \cdot 6 \text{ m}^2$

Question 3

d)



$$\begin{aligned} V &= \pi \int_0^2 (4)^2 dx - \int_0^2 (x^2)^2 dx \\ &= \pi \int_0^2 16 - x^4 dx \\ &= \pi \left[16x - \frac{1}{5}x^5 \right]_0^2 \\ &= \pi \left(32 - \frac{32}{5} \right) - (0 - 0) \\ &= \frac{128\pi}{5} \end{aligned}$$

Volume is $\frac{128\pi}{5}$ units³

Question Four

a)

$$T_n = (x-2)^n$$

$$i) (x-2)^1 + (x-2)^2 + (x-2)^3 + \dots$$

ii)

$$r = (x-2)$$

$$-1 < (x-2) < 1$$

$$1 < x < 3$$

iii)

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{(x-2)}{1-(x-2)}$$

$$= \frac{x-2}{3-x}$$

Question 4

b) (i) $y=0$, $y=e^x-2$
 $0=e^x-2$
 $e^x=2$
 $x=\log_e 2$

(ii) $y=e^x-2$
 $\frac{dy}{dx}=e^x$

$x=1$, $\frac{dy}{dx}=e$

$y-(e-2)=e(x-1)$

$y-e+2=ex-e$

$y=ex-2$

$y=0$,

$0=ex-2$

$ex=2$

$x=\frac{2}{e}$

(iii) $A = \int_{\log_e 2}^1 e^x - 2 \, dx - \int_{\frac{2}{e}}^1 ex - 2 \, dx$

$= \left[e^x - 2x \right]_{\log_e 2}^1 - \left[\frac{ex^2}{2} - 2x \right]_{\frac{2}{e}}^1$

$= (e-2) - (e^{\log_e 2} - 2 \log_e 2) - \left(\left(\frac{e}{2} - 2 \right) - \left(\frac{e}{2} \left(\frac{2}{e} \right)^2 \right) \right)$

$= (e-2 - (2-2 \log_e 2)) - \left(\frac{e}{2} - 2 - \left(\frac{2}{e} - \frac{4}{e} \right) \right)$

$= e - 4 + 2 \log_e 2 - \frac{e}{2} + 2 - \frac{2}{e}$

$= \frac{e}{2} - 2 + 2 \log_e 2 - \frac{2}{e}$

Question Five

a) i) $y' = ax(x-2)$

$(1, -3) \quad -3 = a(-1)$

$$a = 3$$

$$y' = 3x(x-2)$$

$$y' = 3x^2 - 6x$$

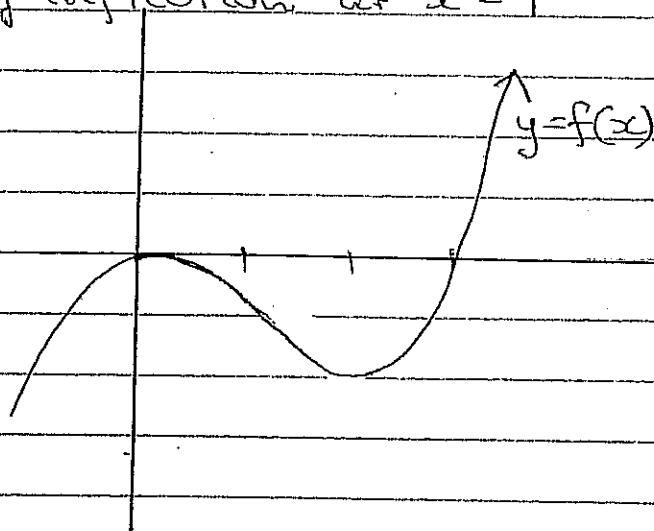
ii) $y = x^3 - 3x^2 + C$
 $= x^2(x-3)$

at $x=0$ $\wedge$ Max t.p.

at $x=2$ $\vee$ Min t.p.

roots at $x=0$ and $x=3$.

Point of inflection at $x=1$



$$b) i) \quad \begin{aligned} \text{1st deposit} & \$1200 \left(1 + \frac{4}{100}\right)^n \\ \text{2nd deposit} & \$600 \left(1 + \frac{4}{100}\right)^{n-1} \end{aligned}$$

$$\text{First 2 deposits} \quad \$1200(1.04)^n + 600(1.04)^{n-1}$$

$$ii) \quad \begin{aligned} \text{3rd deposit amounts to} & 600(1.04)^{n-2} \\ \text{4th deposit amounts to} & 600(1.04)^{n-3} \end{aligned}$$

$$\begin{aligned} \text{Total amount} &= 1200(1.04)^n + 600(1.04)^{n-1} + 600(1.04)^{n-2} + \dots \\ &\quad + \dots + 600(1.04)^2 + 600(1.04) \\ &= 1200(1.04)^n + 600(1.04)(1 + 1.04 + 1.04^2 + \dots + 1.04^{n-1}) \\ &= 1200(1.04)^n + 600(1.04) \left\{ \frac{1(1.04)^n - 1}{1.04 - 1} \right\} \\ &= 1200(1.04)^n + 600(1.04) \frac{(1.04^n - 1)}{0.04} \\ &= 1200(1.04)^n + 15000(1.04)(1.04^n - 1) \\ &= 1200(1.04)^n + 15000(1.04)^n - 15000(1.04) \\ &= 16200(1.04)^n - 15600 \end{aligned}$$

$$(iii) \quad 25000 = 16200(1.04)^n - 15600$$

$$16200(1.04)^n = 40600$$

$$1.04^n = \frac{40600}{16200}$$

$$1.04^n = \frac{203}{81}$$

$$n = \log_{1.04} \left(\frac{203}{81} \right)$$

$$= 23.425 \text{ years}$$

Robyn must make 24 deposits.